



EEG Feature Extraction with Wavelets for Visual Imagery — Concise Guide

1. Task

- Visual Imagery (VI): imagining pictures/scenes without visual input.
- EEG markers subtle vs. evoked responses.
- Key bands/regions:
 - Occipital-parietal alpha (8-12 Hz) ERD/ERS
 - Theta (4-7 Hz), sometimes beta/gamma
- Channels: O1, O2, Oz, POz, PO3/4, Pz (+ frontal controls)

2. Preprocessing

- $F_s \geq 256$ Hz
- Filters: 0.5-40 Hz band-pass, 50/60 Hz notch
- Re-reference: average/mastoids
- Artifact removal: ICA/ASR (blinks/EMG)
- Epochs: -0.2-2.5 s around cue, baseline -200-0 ms
- Standardization: z-score per channel (train stats only)

3. Wavelet Decomposition

- DWT (db4/db6, sym5, coif5): compact subbands → band mapping via levels
- WPD: flexible best-tree subbands

- CWT (Morlet): scalograms for CNNs (time–frequency)
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4. Features

- Power/Energy: absolute & relative bandpower
 - Entropy: Shannon/log-energy
 - Statistics: mean, variance, RMS, skew, kurtosis
 - Hjorth: activity, mobility, complexity
 - Nonlinear: fractal dimension, Hurst, permutation entropy
 - Time–freq: scalogram ridges, maxima
 - Connectivity (opt.): coherence/PLI occipital–parietal
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5. Deep Learning Approaches

- Low-data: EEGNet, ShallowConvNet
 - Image-based: 2D CNN on CWT scalograms
 - Sequence: 1D CNN + attention, TCN, lightweight Transformers
 - Spatial: GNNs, spatial attention
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6. Input Construction

- Windows: 1–2 s, 50% overlap within imagery period
 - Representations:
 - Handcrafted (DWT/WD features → MLP/1D CNN)
 - Raw EEG ([channels × time] → EEGNet/1D CNN/Transformer)
 - CWT scalograms (channels × time × freq → 2D CNN)
 - Normalization: per-channel z-score, optionally per-subject
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7. Training & Validation

- Protocols:
 - Within-subject (simpler)
 - Cross-subject (harder; domain adaptation helps)
- Regularization: dropout, weight decay, early stopping
- Augmentation: Gaussian noise, jitter, warp, amplitude scaling, mixup, SpecAugment, channel dropout

- Imbalance: weighted/focal loss
 - Metrics: accuracy, macro-F1; stratified k-fold / LOSO
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8. Practical Starting Points

- $F_s = 256$ Hz, band-pass 1–40 Hz, notch mains
 - DWT: db4, 5–6 levels $\rightarrow$ energy + entropy + Hjorth
 - CWT: Morlet, 20–40 scales, 4–40 Hz $\rightarrow$ 224×224 scalograms
 - EEGNet: temporal kernel $\approx F_s/2$; depthwise filters = channel groups
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9. Tooling

- Python: MNE, PyWavelets, scipy.signal, PyTorch/Keras, Braindecode
 - MATLAB: EEGLAB, Wavelet Toolbox
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10. Notes

- VI datasets are rare $\rightarrow$ may need custom data collection.
 - Expect lower SNR than VEP/SSVEP; subject-specific models help.
 - Cross-subject generalization is challenging; adapt with normalization/transfer.
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